

EXPLAINER

NATO's Spending Targets Are Measuring the Wrong Thing

Why defence capability cannot be read off a GDP percentage. A systems-based analysis of why NATO's burden-sharing metric optimises for legibility over accuracy, why the Spain case is diagnostic, what Ukraine reveals about the industrial architecture gap, and what this means for Ireland.

Security Ireland

Explainer

IN BRIEF

NATO's burden-sharing debate is framed entirely around financial inputs: what share of GDP each member spends on defence. But defence capability is not a linear function of financial inputs. It is an emergent property of co-evolved systems—technical architectures, industrial bases, and institutional frameworks—that must develop in synchrony before spending converts into actual military capability.

GDP-share targets treat defence as a purchasing exercise. But procurement without the surrounding architecture—interoperable systems, domestic production capacity, trained workforce pipelines, functioning procurement institutions—does not produce capability; it produces expenditure.

The Spain case illustrates the problem precisely: a country can be formally non-compliant on the metric while being strategically correct, because the metric measures inputs rather than outputs. The binding constraint on European defence is not the spending target. It is the set of architectural preconditions that determine whether spending translates into capability at all. This matters for Ireland because its 0.2 per cent of GDP spending is the lowest in the EU—but the coordination architecture argument shifts the debate from metrics Ireland cannot win to structural analysis where Ireland's contribution is relevant.

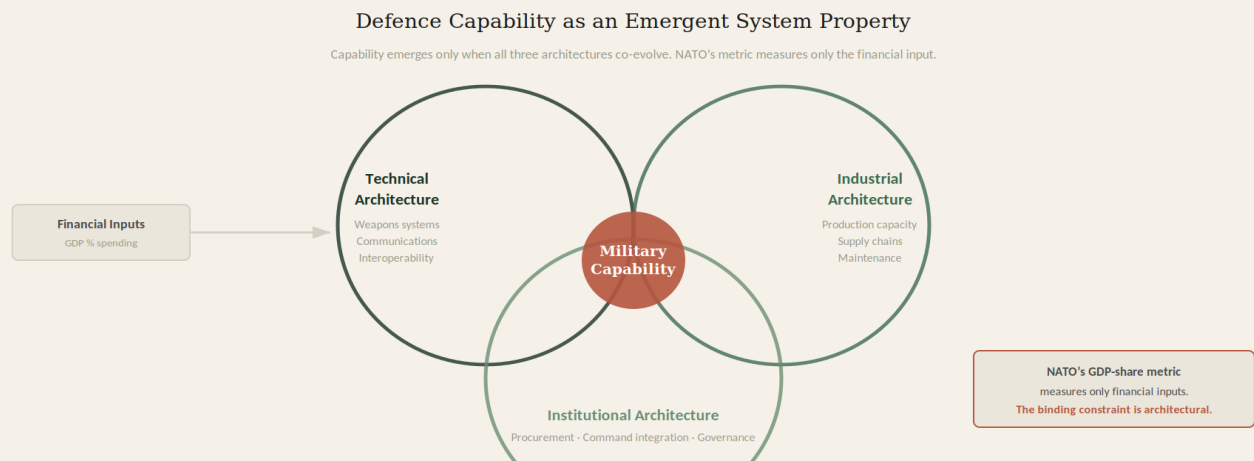


Figure 1: Defence capability as an emergent system property. Financial inputs enable architecture development but do not determine its pace or depth. Military capability emerges only when all three architectures co-evolve in synchrony. NATO's GDP-share metric measures inputs; the binding constraint is architectural co-evolution.

The metric: where 2 per cent came from

NATO's 2 per cent GDP target was never a military doctrine. It originated as a political signal—a burden-sharing shorthand designed to demonstrate commitment to alliance partners. When it was formalised at the Wales Summit in 2014, in the aftermath of Russia's annexation of Crimea, it was explicitly framed as non-binding. NATO's founding treaty contains no GDP-denominated obligation. Member states agree to 'maintain and develop their individual and collective capacity to resist armed attack'—a capabilities commitment, not a financial one.

The target became entrenched for institutional reasons that have little to do with defence strategy. GDP share is legible, comparable, and negotiable across thirty-two

sovereign governments that would never agree on a common capability benchmark. It is a metric optimised for alliance politics, not military effectiveness—an optimisation that is now producing visible costs.

The 5 per cent target now being discussed amplifies the problem. It is not derived from any assessment of what European defence actually requires. It is a number generated by a political demand—that allies demonstrate greater financial commitment to a US-led security architecture—rather than by any analysis of the gap between current and required military capability. Contrary to popular belief, larger inputs fed into an unreformed system do not produce proportionally larger outputs.

Capability is an architecture problem

Defence capability emerges when three types of systems co-evolve: technical architecture (weapons systems, communications, interoperability standards), industrial architecture (production capacity, supply chains, maintenance ecosystems), and institutional architecture (procurement, command integration, burden-sharing governance). Military effectiveness requires all three to function as a coordinated system.

Spending targets address only the financial dimension, and only on the input side. The assumption embedded in them is that if countries spend enough money, the architectures will follow. This assumption has no empirical basis and substantial evidence against it.

Interoperability: the ability of allied forces to communicate, coordinate, and operate together—depends on technical standards that take years to develop and adopt. A NATO member that rapidly scales spending without those standards in place produces platforms that cannot communicate with allied systems.

Industrial capacity: the ability to produce and maintain equipment domestically—takes a decade or more to build from a standing start. A country that surges procurement spending while its industrial base is underdeveloped simply imports that capability from whoever already has the architecture, most commonly the United States.

Workforce capacity: the trained personnel to operate and maintain complex systems—compounds over time and cannot be purchased quickly. No amount of procurement spending creates an experienced military engineering corps in two years.

The result is that defence spending functions as a capability multiplier *only when* the surrounding architectures already exist. Without them, spending is not wasted in the sense of producing nothing—it produces weapons and equipment. But it does not produce the emergent property those weapons and equipment are meant to generate: autonomous, sustainable military capability.

The Spain case as a diagnostic

Spain's refusal to commit to the 5 per cent target has been characterised as free-riding. But Prime Minister Sánchez's actual argument—that committing to 5 per cent 'would not make us any safer' because it 'would only reinforce our dependence'—is architecturally correct, even if it is not articulated in those terms.

In recent years, the more Europe has spent on defence, the more that spending has flowed to American contractors rather than European ones. The European defence industrial base has not scaled proportionally with European defence budgets. What has scaled is European financial dependence on American production architecture—including weapons systems with remote-disable capabilities and supply chains that deprioritise European customers in conflict scenarios.

This is not a side effect of defence spending. It is a predictable consequence of spending without the prior development of industrial architecture. When procurement institutions default to American contractors because European production capacity is insufficient,

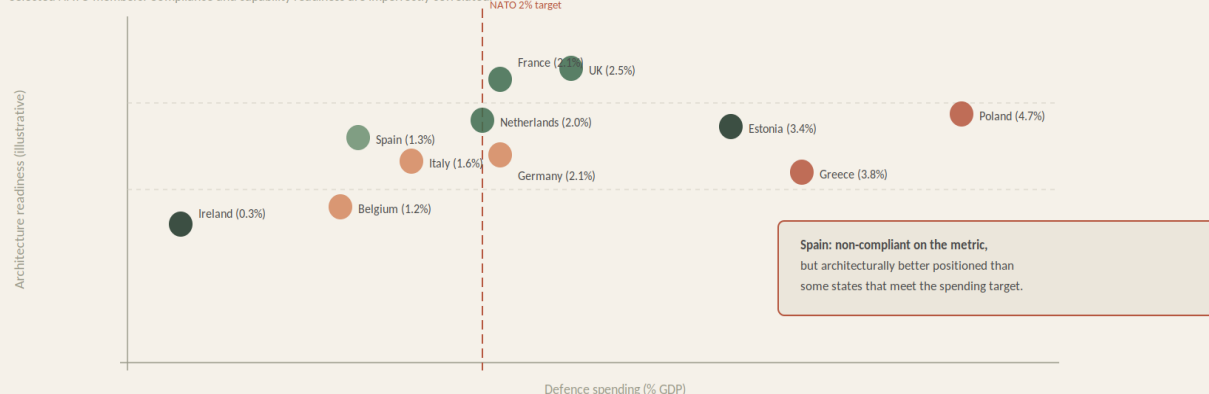
increasing the procurement budget does not solve the structural problem. It amplifies the very outcome it is trying to avoid.

The Spain case is therefore diagnostic in a precise sense. Spain meets the target by routing procurement through external contractors—purchasing American F-35s, for instance—while its own frigate programme and Airbus participation represent genuine industrial architecture investment that the GDP metric does not distinguish. The diagnostic reveals a pattern the GDP-share metric actively conceals: compliance and capability readiness are imperfectly correlated, and some non-compliant states are better positioned architecturally than states that meet the spending target by importing capability.

A country can be formally non-compliant on the GDP metric while being strategically correct, because the metric measures inputs rather than outputs.

Spending Compliance vs Architecture Readiness

Selected NATO members. Compliance and capability readiness are imperfectly correlated.



Source: Security Ireland analysis. Architecture readiness is an illustrative composite index; spending data from NATO 2025 estimates.

Figure 2: Spending compliance versus architecture readiness. The diagnostic reveals a pattern the GDP-share metric actively conceals: compliance and capability readiness are imperfectly correlated.

The Ukraine stress test

The most direct empirical evidence for the architecture argument comes from Ukraine. When the war created sudden, acute demand for European defence production, the industrial architecture gap became impossible to conceal. European defence spending had been rising since 2014. What had not been built, alongside the budgets, was the production architecture to convert that spending into real military output.

The most visible case was artillery ammunition. Ukraine fired more artillery in the first months of the war than Europe's entire pre-war annual production capacity could supply. The EU announced in March 2023 that it would

deliver one million 155mm shells to Ukraine within twelve months: a commitment backed by substantial financial pledges. The actual delivery by March 2024 was approximately 520,000 rounds—barely half the target.

Pre-war European production capacity was approximately 300,000 shells per year, a fraction of Ukraine's peak consumption rate of roughly 3.5 million annually. Even with emergency ramp-up efforts and funding, EU capacity had reached only about 1.4 million rounds per year by end of 2024. The production architecture to absorb even the existing spending was simply not there.

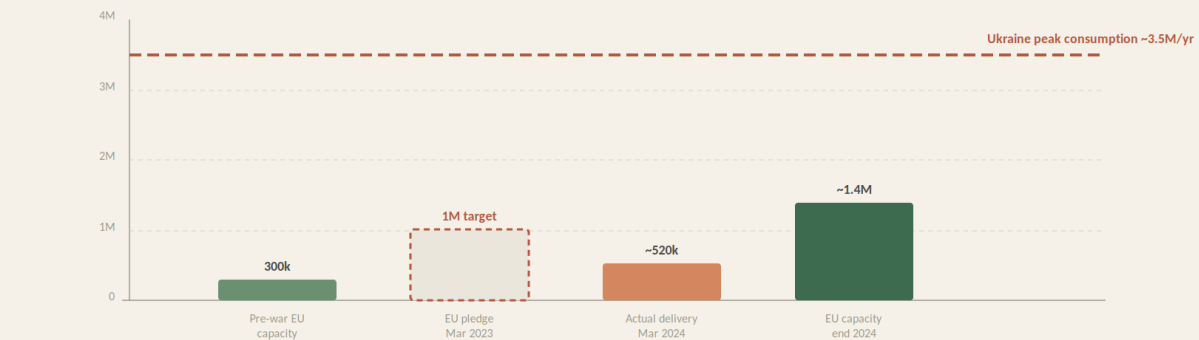
This failure was not financial. European governments had committed the funds. The constraint was industrial: machine tools, skilled workforce, supply chains for propellants and fuses, production lines that take years to build and cannot be conjured by budget announcements. Artillery shells are not such a sophisticated technology that Europe could not produce them at scale. Yet despite years of rising defence budgets, the industrial architecture was absent. This is the clearest possible illustration of what happens when financial commitments flow through

procurement systems that have not built the underlying production architecture.

The same pattern appeared across other categories. Air defence interceptors, armoured vehicles, and anti-tank missiles all faced similar production bottlenecks. In each case, the constraint was not political will or financial resources. It was the industrial architecture that determines whether will and resources convert into output.

The Industrial Architecture Gap: 155mm Artillery Ammunition

European production capacity vs Ukraine wartime consumption, thousands of shells per year



Sources: EDA (2024); IISS; Kiel Institute Ukraine Support Tracker; RUSI (2023). Figures are indicative estimates.

Figure 3: The industrial architecture gap. European ammunition production capacity before the Ukraine war was a fraction of wartime consumption requirements. Emergency ramp-up reached approximately 1.4 million shells per year by end 2024—still below credible deterrence requirements. The bottleneck was never funding; it was production architecture.

What the metric cannot capture

NATO's choice of GDP share as its primary burden-sharing metric reflects an understandable institutional rationale: it is legible, comparable, and politically negotiable, and these properties have real value in a multilateral alliance. The problem is that legibility and strategic validity are different things, and the ease of measurement is not an argument for the validity of what is being measured.

A capability-based framework would ask four different questions:

- (1) Can you produce ammunition and key munitions categories at the scale a sustained conflict would require—or are you dependent on imports that may not arrive when you need them?
- (2) Can your procurement system convert budget authority into signed contracts and delivered equipment within timelines that match the threat environment—or does it take years to spend money that was allocated for urgent capability?
- (3) Can your industrial base sustain production through a conflict without single points of failure in supply chains, workforce, or raw materials—or would a

disruption to one supplier halt production across multiple platforms?

- (4) Can your force structure operate, maintain, and sustain the equipment you procure through a prolonged deployment—or do you buy platforms you cannot crew, arm, or repair?

These are harder to measure than a GDP percentage, because they require joint assessment methodologies, shared data, and institutional willingness to surface uncomfortable findings about structural gaps. They are also the questions that actually determine whether alliance spending converts into collective defence capability. The Ukraine experience has made them impossible to dismiss as theoretical concerns.

NATO has optimised for legibility over accuracy, and the Spain example illustrates the cost. A country that correctly identifies the structural dynamics of European defence dependency—that spending without industrial architecture deepens rather than resolves it—is framed as non-compliant, while countries that hit the spending target by routing procurement through American contractors are framed as fulfilling their obligations.

The architectural precondition

The argument here is not that European defence spending is unimportant. Sustained investment in defence is necessary and overdue. The argument is about what must exist before that spending can do what it is politically intended to do.

Defence capability requires the co-evolution of three architectures: technical, industrial, and institutional. Financial commitments can contribute to that co-evolution if they are sequenced and directed correctly—toward building European production capacity, harmonising procurement institutions, developing common standards,

and funding workforce pipelines. But financial commitments that flow through unreformed procurement systems into existing supply chains do not build those architectures. They entrench the dependency those architectures are meant to replace.

The current NATO spending framework treats the financial commitment as the end point of the burden-sharing problem, but it is not. It is the starting point of an architectural challenge whose binding constraints are not financial.

WHY THIS MATTERS FOR IRELAND

Ireland's defence spending is 0.2 per cent of GDP—the lowest in the EU. In a debate framed entirely around GDP percentages, Ireland has no credible position. But the coordination architecture argument—that Europe's problem is not whether it spends enough but whether twenty-seven procurement systems can convert spending into capability—is uniquely powerful coming from the lowest-spending member state, because it shifts the debate from metrics Ireland cannot win to structural analysis where Ireland's contribution is relevant. Ireland's PESCO participation in seven domain-specific projects (maritime surveillance, subsea protection, cyber resilience) demonstrates the architecture approach in practice: contributing to European security in the domains where Ireland's geographic and institutional position is irreplaceable. During the EU Council Presidency (July–December 2026), Ireland chairs defence budget discussions. The coordination architecture framework gives Ireland an intellectually credible position from which to do so.

The binding constraint on European defence is not the spending target.

It is the architectural precondition that determines whether spending translates into capability at all.

WHY THIS MATTERS NOW

NATO's 5 per cent target is being negotiated as Europe's largest defence spending programme unfolds. If the target is adopted without architectural reform—without building European production capacity, harmonising procurement, and developing common standards—the spending surge will deepen American dependency rather than building European autonomy. The Ukraine ammunition crisis demonstrated this in real time. Ireland's Presidency is the window to make the coordination architecture argument at the European level.

SERIES CONTEXT

This explainer is part of Security Ireland's foundational analysis suite. It connects to *Europe's Defence Problem Isn't Spending* (the coordination architecture applied at EU level), the *Defence Financing Explained* paper (how the money flows), and the *Rearmament Financing* paper (the fiscal constraints). Together they provide the European context for Ireland's domestic security transition.

Security Ireland

Independent analysis for a changing Ireland. · securityireland.ie